Further applications of trigonometry

1

a $x = 7.21$

b $x = 10.00$

c $x = 14.14, y = 14.14$

2

a $x = 4.90$

b $y = 8.66$

c 13.56

3

a $x = 2.70$

b $w = 4.0$

c $z = 3$

4

a $a = 6$

b $b = 6$

c $x = 8.49$

d 36.49 cm

5 $h = 98$

6 $x = 34$

7

a $r = 5 \text{ m}$

b 10 m

8

a $\alpha = 5.96^\circ$

b $\beta = 76.65^\circ$

9

a $x = 21.85$

b $h = 23.95$

10 $L = 13$

11

a 74.85 m

b 28.31°

12 $\theta = 23.9$

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13

a $y = 25.30$ m

b $x = 20.66$ m

c 4.6 m

14

a 16 cm

b 32 cm

c 35.78 cm

15 $h = 4.4$

16

a $x = 5.39$

b $y = 5.39$

c 21 m

d $\$777$

17

a $z = 14.32$

b $\theta = 11.83$

18

a

i $EG = \sqrt{117}$ units

ii $AG = \sqrt{142}$ units

iii $DG = \sqrt{106}$ units

b $3\sqrt{106}$ cm^2

c 59.77°

d $x = 65.19^\circ$

e $y = 49.05^\circ$

19

a 30 cm

b 28.34 cm

c 85.24°

20 32.25°

Additional questions

21

a 11.3 cm

b 12.7 cm

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22

a $x = 34.11$

b To find the area of the parallelogram we first want to find the height of the parallelogram. Notice that a right angle triangle is formed with a hypotenuse of length x and a short leg of $(32 - 11)$ cm. We can find the other leg, let's label it y, of the triangle using either the Pythagorean Theorem, or by solving $\sin 52 = \frac{y}{34}$ for y. Once we have found the perpendicular height of the parallelogram, the area is the product of this height and the base, 32.

c 860 cm^2

23 2532 ft^2

24 58.55°

25 5046 ft^2

26 Any height between 17.4 and 25.9 ft such as 20 ft. We know the length of the hypotenuse and want to find the height of the opposite side to the angle the ramp forms with the ground so we can use the sine ratio. $\sin 10 \approx 0.174$ and $\sin 15 \approx 0.259$

27

a 13.2 cm

b 8.6 cm

28 11.56 cm

29

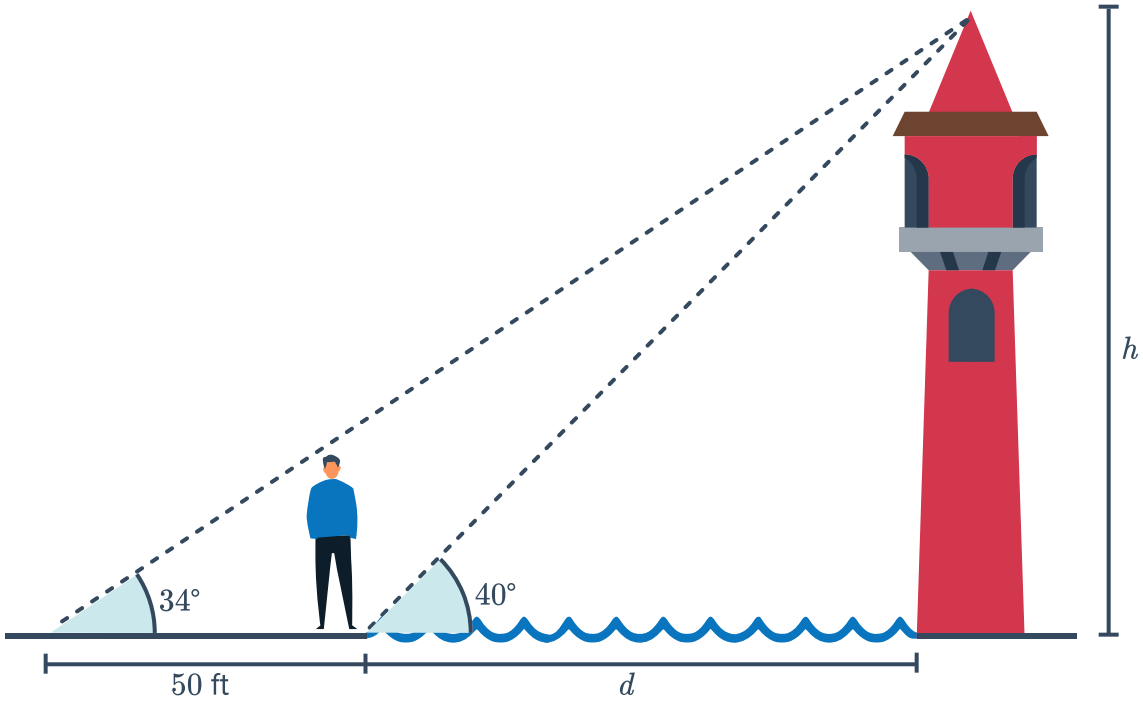
a 271.52 mm

b 211.21 mm

30 946.5 square feet

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31



If we label the distance to the tower from the closer point to be d and the height of the tower to be h we can then use trigonomtry.

For the smaller triangle we get $\tan 40 = \dfrac{h}{d}$, which can be rearranged to give $d \tan 30 = h$.

For the larger triangle we get $\tan 34 = \dfrac{h}{d+50}$, which can rearrange to give $(d+50)\tan 34 = h$.

We can solve for d in the first equation: $d = \dfrac{h}{\tan 40}$, and substituting this into the second equation gives $\left(\dfrac{h}{\tan 30} + 50\right)\tan 34 = h$.

Solving for the height we get $h = 172$ ft

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